

The Measurement Formula Card

Every problem in this chapter is: pick the shape, pick the formula, keep answers exact until the last step.

Shape	Formula	How to think about it
Circle	$C = 2\pi r$; $A = \pi r^2$	arc and sector are FRACTIONS of these: multiply by arc/360
Rhombus / kite	$A = \frac{1}{2} d_1 d_2$	half the product of the diagonals
Regular polygon	$A = \frac{1}{2} a P$	apothem from trig: $a = (s/2) / \tan(180^\circ/n)$
Prism / cylinder	$V = B h$	B is the BASE area; oblique solids too (Cavalieri)
Pyramid / cone	$V = \frac{1}{3} B h$	one third of the matching prism or cylinder
Cone surface	$SA = \pi r^2 + \pi r \ell$	$\ell^2 = r^2 + h^2$ — slant height from a right triangle
Sphere	$SA = 4\pi r^2$; $V = (4/3)\pi r^3$	hemisphere: halve them; add the flat circle πr^2 for SA

The Two Mistakes That Cause Most Lost Points

- Using the diameter as r . Halve the diameter FIRST — πd^2 is not the area of anything.
- Dropping the $\frac{1}{3}$ on pyramids and cones — or using the slant height ℓ in the VOLUME formula. Volume uses the true height h ; only surface area uses ℓ .

Fractions of a Circle, and Solids in Layers

Two pictures organize the whole chapter.

The Fraction Idea (11.1-11.2)

- An arc is $(m/360)$ of the circumference; a sector is $(m/360)$ of the area. Same fraction, different total.
- Backward problems: set up the same equation and solve for the missing piece (angle, radius, or length).
- Radians make it cleaner: $s = r\theta$. Convert with $180^\circ = \pi$ radians.

The Apothem Right Triangle (11.3)

- Split a regular n -gon into n triangles from the center. Half of one triangle is a right triangle.
- Its top angle is half the central angle: $180^\circ/n$. So $\tan(180^\circ/n) = (s/2)/a$.
- Hexagons are special: the triangles are equilateral, so $a = (s/2)\sqrt{3}$ exactly — no calculator.

The Layer Idea (11.4-11.5)

- A prism or cylinder is a stack of identical base layers: $V = (\text{area of layer}) \times (\text{number of layers}) = Bh$.
- Cavalieri: sliding the layers sideways (an oblique solid) changes nothing — same layers, same volume.
- Euler's check for polyhedra: $F + V - E = 2$ catches counting mistakes.

Pointed and Round Solids (11.6-11.8)

- Pointy solids (pyramid, cone) hold exactly $\frac{1}{3}$ of their box or can. Cone slant: $\ell^2 = r^2 + h^2$.
- Sphere: $SA = 4\pi r^2$ (four great circles!) and $V = (4/3)\pi r^3$.
- Composites (silo, capsule): cut into familiar pieces, compute each, add. Keep everything in terms of π until the end.

Name: _____

Which Formula? Practice

Name the formula, then solve. Exact answers unless a decimal is asked for.

1. The length of a 90° arc in a circle of radius 12.
2. The area of a 90° sector in the same circle.
3. The area of a rhombus with diagonals 5 and 12.
4. The exact area of a regular hexagon with side 2.
5. The volume of a cylinder with radius 2 and height 11.
6. The volume of a cone with radius 2 and height 9.
7. The volume of a sphere with radius 3.
8. The surface area of a sphere with radius 3.
9. The slant height of a cone with radius 9 and height 12.
10. The surface area of that cone.
11. The height of a cylinder with radius 3 and volume 63π .
12. The volume of a pyramid with a 7-by-6 base and height 5.
13. The central angle of a regular hexagon, and of a regular decagon.
14. Convert 315° to radians (exact).
15. A capsule: cylinder radius 1, height 4, plus hemispheres on both ends. Total volume?